

Dense Retrieval Pipeline

Page 1 - Retrieval workflow

A dense retrieval pipeline converts technical documents into searchable vectors.

Step 1: Parse Markdown or PDF into structured pages with source metadata.

Step 2: Split each page into stable chunks without crossing PDF page boundaries.

Step 3: Send chunk text to an embedding model and preserve vector order.

Step 4: Upsert chunk IDs, text, metadata, and vectors into persistent Chroma.

Step 5: Embed the user query and perform cosine-distance vector search.

The nearest chunks become retrieval candidates for later answer generation.

Persistence and Metadata

Page 2 - Traceable results

Every chunk uses a stable SHA-256-derived ID so repeated ingestion is idempotent.

PDF chunk metadata retains the original one-based page number and source filename.

Complex fields such as heading paths are serialized safely before Chroma storage.

A new Chroma client can reopen the same persist directory and recover the chunks.

Cosine distance is converted to dense score with: $\text{dense_score} = 1 - \text{distance}$.

Smaller distance therefore produces a larger relevance score.